Weighted Proj bridge: checked homogeneity rung
Pinned Lean 4.34.1; Mathlib d13f23b723b8a846827a245b89c10fc7d3f11612

Weights: X=1, Y=3, Z=1. Coefficients a3,a2,a1,a0 are arbitrary in a field.
Equation: Y^2 - X^5 Z - a3 X^3 Z^3 - a2 X^2 Z^4
          - a1 X Z^5 - a0 Z^6.

CHECKED: equation_weighted_homogeneous, weighted degree six.
CHECKED: relationIdeal_homogeneous, principal ideal homogeneous.
Compiler exit: 0. Both axiom audits: propext, Classical.choice, Quot.sound.
No sorryAx in either audit.

OPEN: grading on the quotient by the homogeneous ideal.
OPEN: degree-zero localization at Z equals the affine coordinate ring.
OPEN: infinity chart and overlap comparison with constructed two-chart curve.
OPEN: smooth projective completion and Pic0 transfer.

The graph and CSV enumerate these obligations without a ranking claim.
Compiler receipt: weighted-proj-head-final-build.log
Copy timeout receipt: weighted-proj-head-copy.txt